Lesson notes · for you

Squeeze and sample

University, vision 3

Materials
A phone with Darsi; Paper for one hand-worked step; The slides from the kit

Learning goals

- Where the brightness sits
- Not the pixels

Activity: In the lab

1. Leave both sliders where they start and read the difference: the two bright specks are already gone.
2. Move Across one step and read the difference again.
3. Say which decode is a new picture, not a repair.

Discussion

- What did the code throw away?

Check yourself

- A nudged code rebuilds, or invents?

Try it this week
Write the two numbers of the original blob and the difference after you move one slider.

Worksheet

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Name _____ Class _____ Date _____

1 **Why does a variational autoencoder keep a cloud of codes for each picture, not one point?**

- ◯ So a new code sampled near it still decodes to a sensible picture
- ◯ So the rebuild keeps every pixel exactly
- ◯ So the code can store the name of the object

2 **Nudging the code gives a new picture; it does not repair the old pixels.**

◯ True ◯ False

3 **Match the model.**

Autoencoder	• • • •	A new code, decoded
Code	• • • •	Squeeze, then rebuild
Sample	• • • •	A short summary

4 **Order one encode and decode.**

- ☐ Store those two numbers
- ☐ Paint a blob at that code
- ☐ Find where the brightness sits

✎ **What did the code fail to keep?**

Check your answers

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1 **So a new code sampled near it still decodes to a sensible picture**

The cloud is what lets you sample: codes in and near it decode to sensible pictures.

2 **True**

It paints a new blob at the new code. The exact pixels were never stored.

3 **Autoencoder — Squeeze, then rebuild**
Code — A short summary
Sample — A new code, decoded

A variational autoencoder (VAE) keeps a cloud of codes, so a new sample can be drawn from it.

4 **Top to bottom: 2, 3, 1**

The picture you get back is the decode, not the file you started with.